

Separation Between Water and Land in SAR Images Using Region-Based Level Sets

Margarida Silveira

Sandra Heleno

IEEE Geoscience and Remote Sensing Letters, 2009

Abstract

This paper presents a semisupervised approach for separating water and land regions in synthetic aperture radar (SAR) amplitude images. The proposed method integrates region-based level set segmentation with probability distribution modeling of SAR pixel intensities. Water and land regions are modeled using finite mixtures of lognormal distributions, whose parameters are estimated using the expectation–maximization (EM) algorithm during contour evolution. Experimental results on real SAR images demonstrate improved robustness and accuracy compared to traditional region-based segmentation approaches.

1 Introduction

Water–land separation in SAR imagery is a fundamental task in environmental monitoring applications such as flood mapping, river monitoring, and coastline extraction. SAR sensors provide all-weather and day-and-night imaging capabilities, which are especially valuable during extreme weather events.

However, SAR image interpretation is challenging due to speckle noise and complex backscattering mechanisms. Water surfaces generally appear as dark regions due to specular reflection, while land surfaces show higher backscatter because of diffuse scattering. Wind roughening, incidence angle variations, and surface heterogeneity can significantly reduce the contrast between water and land regions, making segmentation difficult.

Traditional thresholding and edge-based segmentation methods often fail in such condi-

tions, motivating the use of a region-based statistical segmentation framework.

2 Background and Related Work

Water-land separation can be formulated as a binary classification problem. Early approaches relied on global thresholding after speckle filtering, but these methods are highly sensitive to acquisition conditions. More advanced techniques include Markov Random Fields (MRFs) and active contour models.

Edge-based active contours are unreliable in SAR images due to weak or discontinuous boundaries. Region-based level set methods provide better robustness by using region statistics rather than image gradients. However, many existing region-based methods assume gamma-homogeneous regions, which is an oversimplification for real SAR images where both land and water exhibit heterogeneous scattering characteristics.

3 Proposed Method

The proposed method assumes that the SAR image is partitioned into two regions corresponding to water and land, separated by a closed contour. This contour is represented implicitly using a level set function.

Segmentation is achieved by minimizing an energy functional composed of a data-fitting term and a regularization term. The data-fitting term evaluates how well pixel intensities conform to the statistical models of the two regions, while the regularization term enforces contour smoothness.

To accurately represent SAR image statistics, probability distribution modeling is incorporated into the level set framework.

4 Probability Distribution Modeling

SAR pixel intensities are inherently random due to the coherent nature of radar imaging. Therefore, modeling pixel intensities using probability distributions is more appropriate than relying on raw intensity values.

In this work, both water and land regions are modeled using finite mixtures of lognormal probability density functions. Mixture models allow the representation of heterogeneous

regions composed of multiple scattering mechanisms. Lognormal distributions are particularly suitable for modeling areas with strong spatial variability, such as urban land regions and wind-roughened water surfaces.

This probabilistic modeling provides a flexible and physically meaningful description of SAR amplitude data.

5 Region-Based Level Set Formulation

The segmentation boundary is represented using a level set function that implicitly defines the separation between water and land regions. An energy functional is constructed with two components: a statistical fitting term and a contour regularization term.

The fitting term is derived from the negative log-likelihood of the probability distributions, encouraging pixels inside and outside the contour to match the corresponding region models. The regularization term penalizes excessive boundary complexity by minimizing contour length.

Minimization of this energy functional drives the contour toward the optimal water–land separation.

6 Expectation–Maximization Parameter Estimation

Because the parameters of mixture models cannot be estimated analytically, the expectation–maximization (EM) algorithm is used. During the expectation step, the probability that each pixel belongs to each mixture component is computed. During the maximization step, the parameters of the lognormal distributions and their mixing weights are updated.

Parameter estimation is performed iteratively and concurrently with contour evolution, allowing the statistical model to adapt dynamically as segmentation progresses.

7 Contour Evolution

The evolution of the level set function is governed by a partial differential equation derived from the minimization of the energy functional. The contour motion is driven by the difference between the log-likelihoods of pixel intensities under the water and land models.

A regularized Dirac delta function ensures numerical stability during contour evolution. The level set framework naturally handles changes in topology, such as merging and splitting of regions, which frequently occur in river networks and flooded areas.

8 Experimental Results

The proposed method was evaluated using SAR images acquired by ERS-1, ERS-2, and Envisat ASAR sensors. The datasets include coastal regions, river basins, and flooded areas under varying environmental conditions.

Experimental results show accurate separation of water and land regions, including narrow river channels, sandbanks, and complex flood extents. The method demonstrates strong robustness to wind-roughened water surfaces and heterogeneous land textures.

9 Performance Analysis

Performance was evaluated both qualitatively and quantitatively. Histogram comparisons between observed SAR intensities and estimated probability density functions show correlation coefficients exceeding 97% for both water and land regions.

Quantitative comparisons with gamma-homogeneous level set models were conducted using false positive rate (FPR), false negative rate (FNR), and total classification error (CE). The proposed method consistently achieved lower error rates, particularly in complex flood scenarios.

These results confirm the superiority of lognormal mixture modeling over simpler statistical assumptions.

10 Conclusion

This paper presents a robust method for water–land separation in SAR amplitude images by combining region-based level sets with probability distribution modeling. The key contribution lies in the use of finite mixtures of lognormal distributions estimated via the EM algorithm during contour evolution.

The proposed framework effectively handles speckle noise, texture variability, and weak boundaries. Although demonstrated for water–land separation, the methodology is gen-

eral and can be extended to other SAR image segmentation tasks such as oil spill detection and ice–water separation.